NATIONAL UNIVERSITY OF SINGAPORE

CS5224 – Cloud Computing

(Semester 2: AY 2024/2025)

Time Allowed: 1.5 Hours

Sample Answers

INSTRUCTIONS TO STUDENTS

1. Please write your Student Number only in the answer sheet. **Do not** write your name.

2. This assessment paper contains **TWENTY FIVE (25)** questions and comprises **EIGHT (8)** printed pages (including this cover page).

3. Total marks in this question paper is **30**.

4. Students are required to answer **ALL** questions.

5. Write your answers in the space provided **ONLY** on the answer sheet.

6. This is an **OPEN BOOK** assessment.

7. Only printed/written materials are allowed. Electronic devices are not allowed.

Examiner's Use Only

Question No.	Maximum Marks
MCQs 1 – 20	20
21	2
22	2
23	2
24	2
25	2
Total	30

Question 1

Which one of the following would be the best use case for SaaS?

(a) Applications with highly confidential and sensitive data
(b) A mobile application
(c) An application used by an organisation with people working around the world, where some have erratic or unreliable internet connections
(d) Organisations looking to reduce on-premises IT infrastructure and capital expenditure
(e) Applications that need a lot of user data

Option (d). For the rest, SaaS is not a practical application to use.

Question 2

Which of the following statements is false?

(a) Containers and virtual machines have similar resource isolation and distribution advantages
(b) Elastic load balancing automatically distributes access traffic to multiple cloud servers, expands the external service capabilities of application systems and achieves higher levels of application fault tolerance
(c) Moving a SaaS deployment from one cloud provider to another is an easy proposition
(d) Hypervisors separate VMs logically and assigns each a share of the underlying computing power, memory, and storage
(e) The NIST definition of cloud computing includes five essential characteristics, four deployment models, and three service models

Option (c). It is not easy as there may be many elements of interoperability, not least with the infrastructure.

Question 3

Which level of cloud service is usually provided for the application to run the development environment?

(a) PaaS
(b) IaaS
(c) SaaS
(d) FaaS
(e) None of the above

Option (a).

Question 4

What cloud benefits make it lower risk for enterprises to adopt the cloud?

(a) The pay-as-you-go model allows enterprises to experiment with technologies as opposed to making long-term decisions based on little or no trial
(b) Data security associated with loss or unavailability of data causing business disruption
(c) The advantage of not managing infrastructure
(d) Standardized APIs and integration tools simplify interoperability, reducing complexity and adoption risks
(e) None of the above

Option (a). (b) and (d) are actually issues. (c) is only a benefit for some types of cloud deployments and not all.

Question 5

Is it possible to run completely different operating systems on virtual machines (VMs) hosted on a single physical host with an OS? If yes, what makes this possible?

(a) Yes, virtualization makes it possible—VMs can run different operating systems on a single host
(b) No, it is not possible – VMs are software-based versions of a single host. They cannot have different environments from one another
(c) Yes, containerization makes it possible – to have VMs that are unique physical entities, so you can have completely different operating systems on them
(d) No, it is not possible – A single host can only work as one single virtual environment, and can, therefore, have only one environment
(e) Yes, it is possible – this can be done through a Type 1 Hypervisor

Option (a). Containers share host OS, while a Type 1 hypervisor is directly on H/W

Question 6

A company which recently migrated to the cloud is noticing that its customers are facing slow data access times. What could be the main cause?

(a) Virtualisation
(b) Dedicated server allocation
(c) Elastic load balancing
(d) Automatic scaling
(e) Low Bandwidth

Option (e). The rest actually help alleviate slowness in some ways.

Question 7

Which of the following is an example of Software as a Service (SaaS)?

(a) Microsoft Azure
(b) Amazon EC2
(c) Google Cloud Platform
(d) Google Photos
(e) Amazon EMR

Option (d). Rest are all cloud provisions – either PaaS or IaaS.

Question 8

An organisation *O* (cloud service consumer), accesses a cloud service offered by public cloud *A*. Based on this, which of the following statements is false?

(a) *O* has its own organisational boundary
(b) *A* has its own organisational boundary
(c) *O* extends its trust boundary to encompass the organisational boundary of *A*
(d) *A* extends its trust boundary to encompass the organisational boundary of *O*
(e) One of the above statements is false

Option (d). Both *O* and *A* define their organisational boundaries, but cloud consumer must also extend its boundary to trust the cloud provider, and not the other way around.

Question 9

Which of the following scenarios is best suited for using a hybrid cloud model?

(a) A government agency hosting sensitive data
(b) A startup hosting a website
(c) A user hosting a blog
(d) A large enterprise with varying workloads
(e) A startup having a lot of data

Option (d). Hybrid clouds allow balancing between private and public clouds.

Question 10

Your company uses virtual machines (VMs) and is experiencing resource contention (e.g., CPU, memory, disk I/O, or network). Which solution is most effective to address this issue comprehensively?

(a) Add more physical servers
(b) Add more memory
(c) Use faster storage devices
(d) Increase your network speeds
(e) Optimise VM resource allocation

Option (e). Adding more physical servers may help but it depends more on how the virtualisation is.

Question 11

A company provides users with a SaaS to file their taxes. Close to the deadline, there is a huge spike in demand. Which strategy is most effective for ensuring performance and availability for this SaaS during the spike?

(a) Design a robust load balancing architecture
(b) Design a system for the highest probable peak load
(c) Upgrade network connection to deal with potential spike
(d) Lease additional hardware during peak period, as necessary
(e) Ask the users to wait in an online queue

Option (a). We can never estimate the real peak and leasing expensive hardware is not an option due to delays in provisioning. (c) and (e) are not really options.

Question 12

What is the advantage of a multitenancy cloud environment over a single tenancy environment?

(a) Easy to customize
(b) Cost savings
(c) Faster performance
(d) Higher data security
(e) Higher data governance

Option (b). Multiple users share the same infrastructure and resource, thus leading to better resource utilisation and hence cost savings.

Question 13

Your company has built a SaaS that is meant to serve around 5000 customers. However, your SaaS has become very popular (yay!) and suddenly there is a demand from about 25000 users. This will lead to an increase in resource utilisation. What are the incremental maintenance costs (mainly management overheads such as configuring servers, patching, etc.) of adding these new resources?

(a) About the same as before
(b) Twice the original cost
(c) Three times the original cost
(d) Four times the original cost
(e) Five times the original cost

Option (a). Not much extra maintenance since the extra resources come from the cloud, which is managed by the cloud provider. There are likely extra running costs.

Question 14

What is the advantage of a single tenancy cloud environment over a multitenancy environment?

(a) Lower costs due to shared resources
(b) Enhanced data security and isolation
(c) Faster scalability for user spikes
(d) Simplified resource management
(e) Reduced need for compliance controls

Option (b).

Question 15

What is one key advantage of using virtual machines over containers?

(a) Requires less memory space
(b) More secure
(c) Faster startup time
(d) Easy to scale
(e) More cost-effective

Option (b). Each VM is isolated, while containers can be impacted by OS being compromised.

Question 16

Which of the following statements about cloud computing is always true?

(a) It cannot be used in environments where data needs to be stored within a particular region/boundary
(b) It helps protect your data through encryption
(c) It enables users to access systems and resources regardless of location
(d) It introduces latency as servers are geographically distributed
(e) It can improve web server response time by having servers closer to the user

Option (c). For (a), you can use a local provider or a sovereign cloud. (b) only in certain cases based on the cloud provider and your settings. (d) is not correct, as latency is only an issue in some cases in this scenario. (e) may or may not help.

Question 17

Your company operates a multi-cloud environment (e.g., AWS and Azure) and is experiencing data synchronization issues between the clouds, such as incomplete transfers or data loss. What is the most likely cause?

(a) Slow storage performance on one cloud
(b) Limited virtualisation on one cloud
(c) Incompatible cloud APIs
(d) Uneven resource provisioning (e.g., storage, compute, or network)
(e) Inconsistent security configurations

Option (d). You could end up with less space for the data on one of the clouds.

Question 18

Out of the following five compute models, which one requires the most maintenance?

(a) PaaS and VMs
(b) FaaS and Containers
(c) PaaS and Containers
(d) Containers and VMs
(e) PaaS and FaaS

Option (d). PaaS is managed by the provider, while FaaS requires low maintenance.

Question 19

You wish to migrate to the cloud to run a simple server side computation which maintains no state about clients. Which of the following is best suited for this purpose?

(a) Amazon EC2
(b) REST
(c) SOAP
(d) AWS Lambda
(e) Amazon EMR

Option (d). Use FaaS – computations are simple, and no state is maintained.

Question 20

What is a primary challenge when migrating legacy on-premise systems to the cloud?

(a) Decreased performance
(b) Increased data compliance
(c) Increased management
(d) Increased costs
(e) Application compatibility

Option (e). (b) is not an issue depending on choosing the right cloud.

Question 21

You are a cloud developer who is deploying your application (which has a steady-state workload) to AWS. Which pricing option would you choose and why?

Answer: You will choose the reserved option as this is typically used to run 24/7 over the long term and the cost is lower than on-demand.

Question 22

The company you work for is responsible for a large web server farm. You are tasked with increasing the availability of this farm to ensure little to no downtime. List two ways in which you can achieve this.

Answer: You want to ensure high probability that your system is operational at a given time. You can use scaling (auto-scaling in particular and have redundancy, where your server instances are in multiple zones/locations.

Question 23

Your company is using a public cloud environment but is facing some security concerns regarding its data. As the lead cloud engineer, you are asked to come up with the solution, while keeping costs down. How would you do it? Explain your answer.

Answer: Moving to a hybrid cloud will help as sensitive data can be stored on a private cloud and less sensitive workloads can be performed. You can also move everything to a private cloud but that would be more expensive overall.

Question 24

The time it takes to back up your data on the cloud depends only on the amount of data you have. Is this statement true or false? Explain your answer

Answer: False. Performance = Function of (Server Location, Latency, Bandwidth, Data Amount).

Question 25

Your workplace wishes to make the work environment a bit more fun ☺. They are going to provide advanced collaboration services, such as video, chat, and an online whiteboard. Given the cost of hosting the infrastructure for this, they have decided that a cloud-based approach is the way to go. Which cloud computing deployment* model would best suit your company's needs? Explain your answer.

Answer: It depends! ☺. The main issue is data security and privacy and whether you require it or not. If it is not as required, you would go with a public cloud and if it were, a hybrid or private model would work fine.
*corrected during the test

– END OF PAPER –